2022 AIME I Problems/Problem 12

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Problem

For any finite set X, let $|X|$ denote the number of elements in X. Define

$$S_n = \sum |A \cap B|,$$

where the sum is taken over all ordered pairs (A, B) such that A and B are subsets of $\{1, 2, 3, \cdots, n\}$ with $|A| = |B|$. For example, $S_2 = 4$ because the sum is taken over the pairs of subsets

$$(A, B) \in \{(\emptyset, \emptyset), (\{1\}, \{1\}), (\{1\}, \{2\}), (\{2\}, \{1\}), (\{2\}, \{2\}), (\{1, 2\}, \{1, 2\})\},$$

giving $S_2 = 0 + 1 + 0 + 0 + 1 + 2 = 4$. Let $\frac{S_{2022}}{S_{2021}} = \frac{p}{q}$, where p and q are relatively prime positive integers. Find the remainder when $p + q$ is divided by 1000.

Solution 1 (Easy to Understand)

Let's try out for small values of n to get a feel for the problem. When $n = 1, S_n$ is obviously 1. The problem states that for $n = 2, S_n$ is 4. Let's try it out for $n = 3$.

Let's perform casework on the number of elements in A, B.

$\textbf{Case 1:} |A| = |B| = 1$

In this case, the only possible equivalencies will be if they are the exact same element, which happens 3 times.

$\textbf{Case 2:} |A| = |B| = 2$

In this case, if they share both elements, which happens 3 times, we will get 2 for each time, and if they share only one element, which also happens 6 times, we will get 1 for each time, for a total of 12 for this case.

$\textbf{Case 3:} |A| = |B| = 3$

In this case, the only possible scenario is that they both are the set $\{1, 2, 3\}$, and we have 3 for this case.

In total, $S_3 = 18$.

Now notice, the number of intersections by each element $1 \ldots 3$, or in general, $1 \ldots n$ is equal for each element because of symmetry - each element when $n = 3$ adds 6 to the answer. Notice that $6 = \binom{4}{2}$ - let's prove that $S_n = n \cdot \binom{2n-2}{n-1}$ (note that you can assume this and answer the problem if you're running short on time in the real test).

Let's analyze the element k - to find a general solution, we must count the number of these subsets that k appears in. For k to be in both A and B, we need both sets to contain k and another subset of 1 through n not including k. ($A = \{k\} \cup A' | A' \subset \{1, 2, \ldots, n\} \land A' \not\subset \{k\}$ and $B = \{k\} \cup B' | B' \subset \{1, 2, \ldots, n\} \land B' \not\subset \{k\}$)

For any $0 \leq l \leq n - 1$ that is the size of both A' and B', the number of ways to choose the subsets A' and B' is $\binom{n-1}{l}$ for both subsets, so the total number of ways to choose the subsets are $\binom{n-1}{l}^2$. Now we sum this over all possible l's to find the total number of ways to form sets A and B that contain k. This is equal to $\sum_{l=0}^{n-1}\binom{n-1}{l}^2$. This is a simplification of Vandermonde's identity, which states that $\sum_{k=0}^{r}\binom{m}{k}\cdot\binom{n}{r-k}=\binom{m+n}{r}$. Here, m, n and r are all $n-1$, so this sum is equal to $\binom{2n-2}{n-1}$. Finally, since we are iterating over all k's for n values of k, we have $S_n=n\cdot\binom{2n-2}{n-1}$, proving our claim.

We now plug in S_n to the expression we want to find. This turns out to be $\frac{2022\cdot\binom{4042}{2021}}{2021\cdot\binom{4040}{2020}}$. Expanding produces $\frac{2022\cdot 4042!\cdot 2020!\cdot 2020!}{2021\cdot 4040!\cdot 2021!\cdot 2021!}$.

After cancellation, we have

$$\frac{2022\cdot 4042\cdot 4041}{2021\cdot 2021\cdot 2021}\implies\frac{4044\cdot 4041}{2021\cdot 2021}$$

4044 and 4041 don't have any common factors with 2021, so we're done with the simplification. We want to find $4044\cdot 4041+2021^2\pmod{1000}\equiv 44\cdot 41+21^2\pmod{1000}\equiv 1804+441\pmod{1000}\equiv 2245\pmod{1000}\equiv\boxed{245}$

~KingRavi ~Edited by MY-2

Solution 2 (Linearity of Expectation)

We take cases based on the number of values in each of the subsets in the pair. Suppose we have k elements in each of the subsets in a pair (for a total of n elements in the set). The expected number of elements in any random pair will be $n\cdot\frac{k}{n}\cdot\frac{k}{n}$ by linearity of expectation because for each of the n elements, there is a $\frac{k}{n}$ probability that the element will be chosen. To find the sum over all such values, we multiply this quantity by $\binom{n}{k}^2$. Summing, we get

$$\sum_{k=1}^{n}\frac{k^2}{n}\binom{n}{k}^2$$

Notice that we can rewrite this as

$$\sum_{k=1}^{n}\frac{1}{n}\left(\frac{k\cdot n!}{(k)!(n-k)!}\right)^2=\sum_{k=1}^{n}\frac{1}{n}n^2\left(\frac{(n-1)!}{(k-1)!(n-k)!}\right)^2=n\sum_{k=1}^{n}\binom{n-1}{k-1}^2=n\sum_{k=1}^{n}\binom{n-1}{k-1}\binom{n-1}{n-k}$$

We can simplify this using Vandermonde's identity to get $n\binom{2n-2}{n-1}$. Evaluating this for 2022 and 2021 gives

$$\frac{2022\binom{4042}{2021}}{2021\binom{4040}{2020}}=\frac{2022\cdot 4042\cdot 4041}{2021^3}=\frac{2022\cdot 2\cdot 4041}{2021^2}$$

Evaluating the numerators and denominators mod 1000 gives $804+441=1\boxed{245}$

- pi_is_3.14

Solution 3 (Rigorous)

For each element i, denote $x_i=(x_{i,A},x_{i,B})\in\{0,1\}^2$, where $x_{i,A}=\mathbb{I}\{i\in A\}$ (resp. $x_{i,B}=\mathbb{I}\{i\in B\}$).

Denote $\Omega=\left\{(x_1,\cdots,x_n):\sum_{i=1}^{n}x_{i,A}=\sum_{i=1}^{n}x_{i,B}\right\}$.

Denote $\Omega_{-j}=\left\{(x_1,\cdots,x_{j-1},x_{j+1},\cdots,x_n):\sum_{i\neq j}x_{i,A}=\sum_{i\neq j}x_{i,B}\right\}$.

Hence,

$$\begin{aligned}
S_n &= \sum_{(x_1,\cdots,x_n)\in\Omega} \sum_{i=1}^{n} \mathbb{I}\left\{x_{i,A} = x_{i,B} = 1\right\} \\
&= \sum_{i=1}^{n} \sum_{(x_1,\cdots,x_n)\in\Omega} \mathbb{I}\left\{x_{i,A} = x_{i,B} = 1\right\} \\
&= \sum_{i=1}^{n} \sum_{(x_1,\cdots,x_{i-1},x_{i+1},\cdots,x_n)\in\Omega_{-i}} 1 \\
&= \sum_{i=1}^{n} \sum_{j=0}^{n-1} \left(\binom{n-1}{j}\right)^2 \\
&= n \sum_{j=0}^{n-1} \left(\binom{n-1}{j}\right)^2 \\
&= n \sum_{j=0}^{n-1} \binom{n-1}{j}\binom{n-1}{n-1-j} \\
&= n\binom{2n-2}{n-1}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{S_{2022}}{S_{2021}} &= \frac{2022\binom{4042}{2021}}{2021\binom{4040}{2020}} \\
&= \frac{4044 \cdot 4041}{2021^2}.
\end{aligned}$$

This is in the lowest term. Therefore, modulo 1000,

$$\begin{aligned}
p + q &\equiv 4044 \cdot 4041 + 2021^2 \\
&\equiv 44 \cdot 41 + 21^2 \\
&\equiv \boxed{\textbf{(245)}}.
\end{aligned}$$

~Steven Chen (www.professorchenedu.com)

Solution 4

Let's ask what the contribution of an element $k \in \{1, 2, \cdots, n\}$ is to the sum $S_n = \sum |A \cap B|$.

The answer is given by the number of (A, B) such that $|A| = |B|$ and $k \in A \cap B$, which is given by $\binom{2n-2}{n-1}$ by the following construction: Write down 1 to n except k in a row. Do the same in a second row. Then choose $n-1$ numbers out of these $2n-2$ numbers. k and the numbers chosen in the first row make up A. k and the numbers not chosen in the second row make up B. This is a one-to-one correspondence between (A, B) and the ways to choose $n-1$ numbers from $2n-2$ numbers.

The contribution from all elements is therefore

$$S_n = n\binom{2n-2}{n-1}.$$

For the rest please see Solution 1 or 2.

~qyang

NOTE: you may have a concern, "what if we chose the same element twice in a set?" Let us look at an example. Suppose we are partioning $S_4,\ k = 4,$ and we list out the rows top: $1, 2, 3$ bottom: $1, 2, 3$ If we chose 1 (top) 2 (top) and 1 (bottom), then we can distribute these numbers by sending the numbers in the top row that were chosen to set $A,$ which would be $1, 2, 4,$ and set B would be the numbers not chosen in the bottom row, which would be $2, 3, 4.$ To prove this rigourously, let x be the number of elements in the top row chosen and y be the number of elements chosen in the bottom row. Since we choose $n-1$ objects, $x + y = n - 1.$ So the amount of elements in A would be $1 + x,$ and the amount of elements in B would be

$$1+(n-1-y)=1+n-1-(n-1-x)=x+1.$$

~eqb5000

Solution 5 (No strange notation, proof of all combo lemmas provided)

Assume the subset A of $\{1,2,3,\ldots,n\}$ has k elements, where k is some integer from 0 to n, inclusive. Then, there are $\binom{n}{k}$ ways to choose A. Now, we find the amount of ways to choose B, given A. Assume that A and B have a elements in common, where $0 \leq a \leq k$. Then, a of the elements in B are also in A, so we can pick $\binom{k}{a}$ elements here, and the remaining $k-a$ elements in B are $\underline{\text{not}}$ in A, so we can pick $\binom{n-k}{k-a}$ elements here. Therefore, there are $\binom{k}{a}\binom{n-k}{k-a}$ ways to select B if A and B have a elements in common.

However, if A and B have a elements in common, the ordered pair (A,B) will contribute a to the total sum. Thus, we multiply by a. Since the amount of ways to choose B ranges over all $0 \leq a \leq k$, we seek the sum;

$$\binom{n}{k}\sum_{a=0}^{k} a\binom{k}{a}\binom{n-k}{k-a}.$$

The $\binom{n}{k}$ on the outside of the sum represents the amount of ways to choose A. Now, we simplify the $a\binom{k}{a}$ term a little bit

Note that $a\binom{k}{a}=a\cdot\dfrac{k!}{a!(k-a)!}=\dfrac{k!}{(a-1)!(k-a)!}=k\cdot\dfrac{(k-1)!}{(a-1)!(k-a)!}=k\binom{k-1}{a-1}$

The reason we did this is because we want the a in the combination expression, and not as a scale factor. Then, the sum simplifies to

$\binom{n}{k}\sum_{a=0}^{k} k\binom{k-1}{a-1}\binom{n-k}{k-a}$. We can take k out of the sum (it's a constant in the sum), and the sum turns into

$$k\binom{n}{k}\sum_{a=0}^{k}\binom{k-1}{a-1}\binom{n-k}{k-a}.$$

Now, we use a combinatorial identity called Vandermode's Identity (This identity is well known, if you don't know it, learn it; it's useful for these types of problems). Consider $x+y$ people. We wish to form a committee with r people. We can do this in $\binom{x+y}{r}$ ways. However, we can choose k people from the group of x people and $r-k$ people from the other y people, for all $0 \leq k \leq r$.

This is done in $\sum_{k=0}^{r}\binom{x}{k}\binom{y}{r-k}$ ways, but this is also equal to $\binom{x+y}{r}$.

In our desired sum, we have $x=k-1$, $y=n-k$, and $r=(a-1)+(k-a)=k-1$. Thus, the summation part is equal to $\binom{n-1}{k-1}$, and the total product is $k\binom{n}{k}\binom{n-1}{k-1}$.

Now, we want to sum this over all possible k, for $0 \leq k \leq n$.

This is $\sum_{k=0}^{n} k\binom{n}{k}\binom{n-1}{k-1}$. We can do something with the $k\binom{n}{k}$ term that was similar to something we did last time to obtain that this is also equal to $n\binom{n-1}{k-1}$.

Then, the sum simplifies to $n\sum_{k=0}^{n}\binom{n-1}{k-1}\binom{n-1}{k-1}=n\sum_{k=0}^{n}\binom{n-1}{k-1}^2$.

We can use Vandermode's again, with $x=y=n-1$ and $r=(k-1)+(k-1)=2k-2$. Then, the sum is equal to $n\binom{2n-2}{n-1}$.

Now, the problem seeks $\dfrac{S_{2022}}{S_{2021}}$. This is equal to

$$\frac{2022\binom{4042}{2021}}{2021\binom{4040}{2020}} = \frac{\frac{2022\cdot 4042!}{2021!\cdot 2021!}}{\frac{2021\cdot 4040!}{2020!\cdot 2020!}} = \frac{2022\cdot 4042!\cdot 2020!\cdot 2020!}{2021\cdot 4040!\cdot 2021!\cdot 2021!} = \frac{2022\cdot 4042\cdot 4041}{2021\cdot 2021\cdot 2021} = \frac{2022\cdot 2\cdot 4041}{2021\cdot 2021}.$$

We seek $2022\cdot 2\cdot 4041 + 2021\cdot 2021 \pmod{1000}$. This is equivalent to $22\cdot 82 + 21\cdot 21 \equiv 804 + 441 \equiv \boxed{245} \pmod{1000}$, and we are done.

Video Solution

https://youtu.be/cXJmHV5BnfY ~MathProblemSolvingSkills.com

https://youtu.be/wTYXkE32v9o ~AMC & AIME Training

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